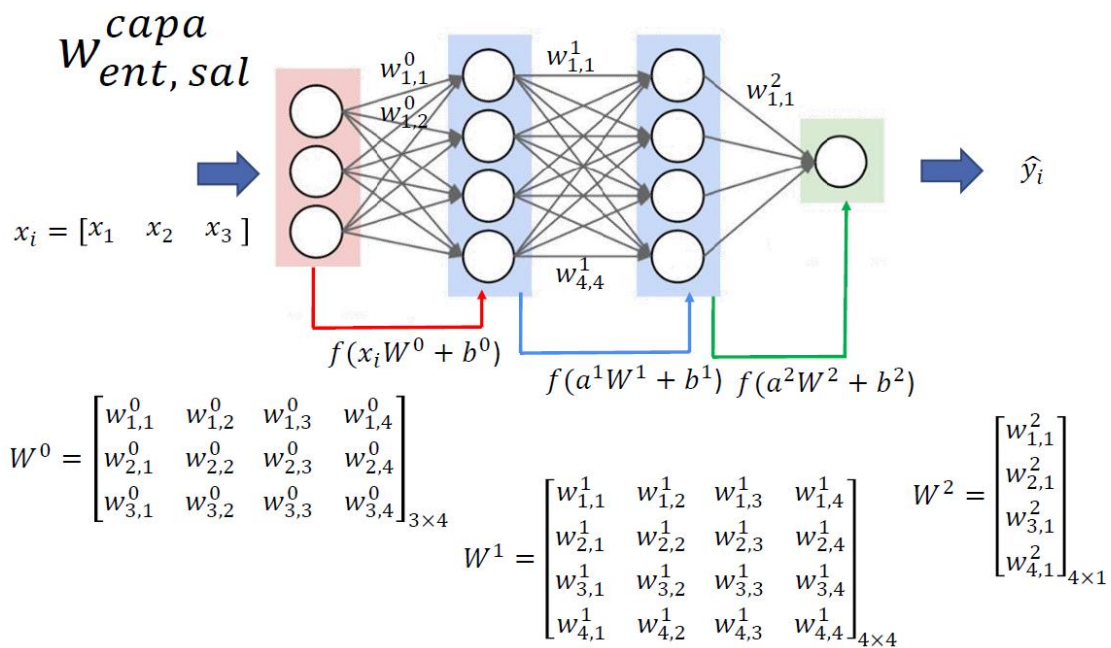
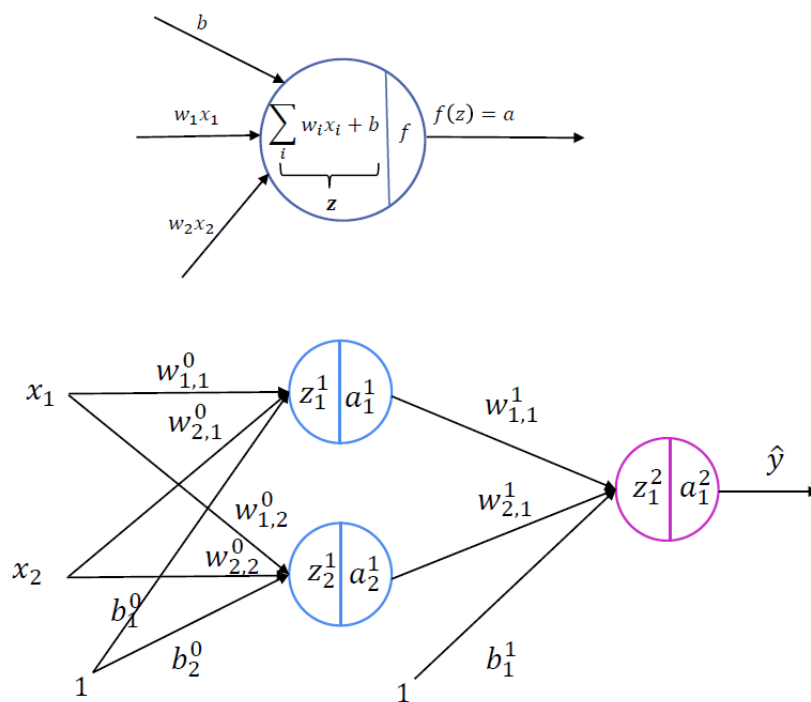


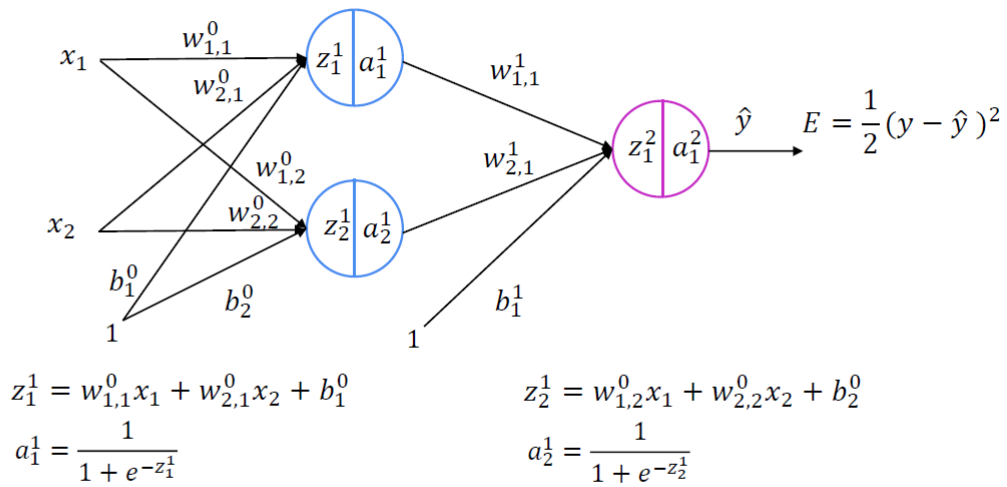
Estructura de una red neuronal



Forward and Backward propagation



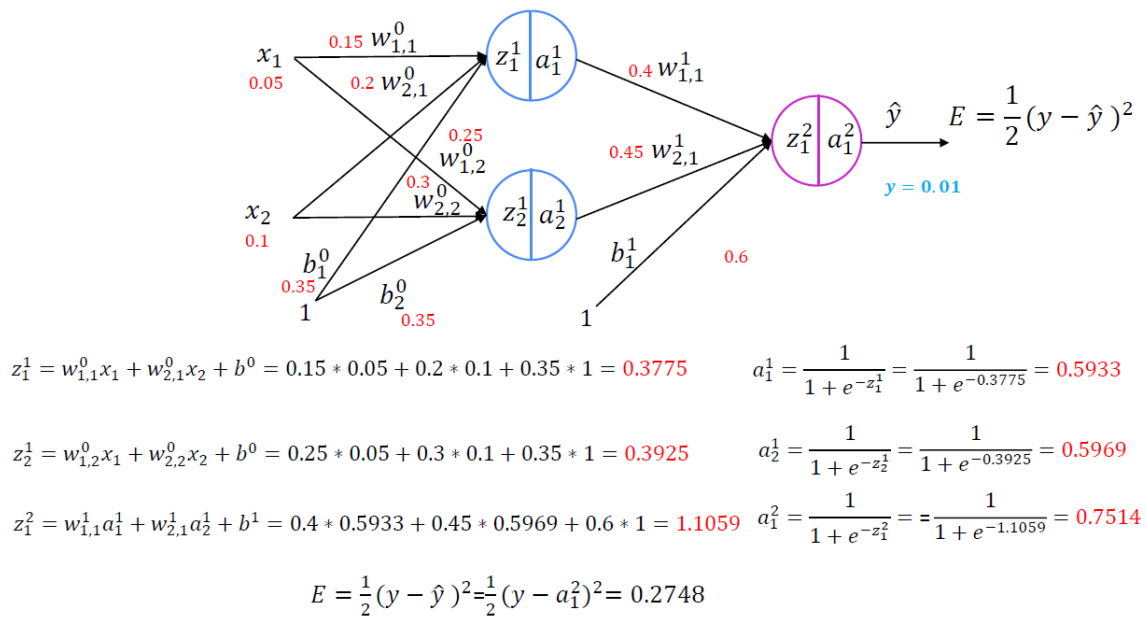
Forward



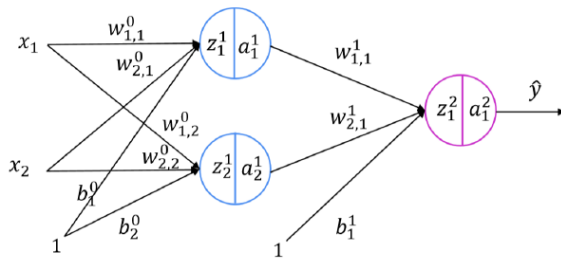
$$z_1^2 = w_{1,1}^1 a_1^1 + w_{2,1}^1 a_2^1 + b_1^1$$

$$a_1^2 = \frac{1}{1 + e^{-z_1^2}} = \hat{y}$$

Forward propagation



Backpropagation



1. Actualización de pesos etapa 1

$$w_{1,1}^1(t+1) = w_{1,1}^1(t) - \eta \frac{\partial E}{\partial w_{1,1}^1}$$

$$w_{2,1}^1(t+1) = w_{2,1}^1(t) - \eta \frac{\partial E}{\partial w_{2,1}^1}$$

1. Actualización de pesos etapa 0

$$w_{1,1}^0(t+1) = w_{1,1}^0(t) - \eta \frac{\partial E}{\partial w_{1,1}^0}$$

$$w_{2,1}^0(t+1) = w_{2,1}^0(t) - \eta \frac{\partial E}{\partial w_{2,1}^0}$$

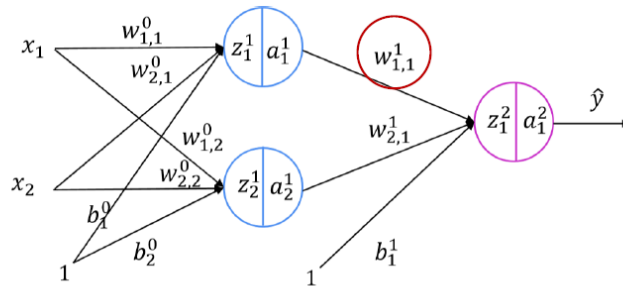
$$w_{1,2}^0(t+1) = w_{1,2}^0(t) - \eta \frac{\partial E}{\partial w_{1,2}^0}$$

$$w_{2,2}^0(t+1) = w_{2,2}^0(t) - \eta \frac{\partial E}{\partial w_{2,2}^0}$$

Backpropagation

$$w_{1,1}^1(t+1) = w_{1,1}^1(t) - \eta \frac{\partial E}{\partial w_{1,1}^1}$$

$$\frac{\partial E}{\partial w_{1,1}^1}$$



$$E = \frac{1}{2}(y - \hat{y})^2 = \frac{1}{2}(y - a_1^2)^2$$

$$\frac{\partial E}{\partial w_{1,1}^1} \longrightarrow a_1^2 = \frac{1}{1 + e^{-z_1^2}} \longrightarrow z_1^2 = w_{1,1}^1 a_1^1 + w_{2,1}^1 a_2^1 + b_1^1$$



Regla de la cadena

$$\frac{\partial E}{\partial w_{1,1}^1} = \frac{\partial E}{\partial a_1^2} * \frac{\partial a_1^2}{\partial z_1^2} * \frac{\partial z_1^2}{\partial w_{1,1}^1}$$

Backpropagation

$$\frac{\partial E}{\partial w_{1,1}^1} = \frac{\partial E}{\partial a_1^2} * \frac{\partial a_1^2}{\partial z_1^2} * \frac{\partial z_1^2}{\partial w_{1,1}^1}$$

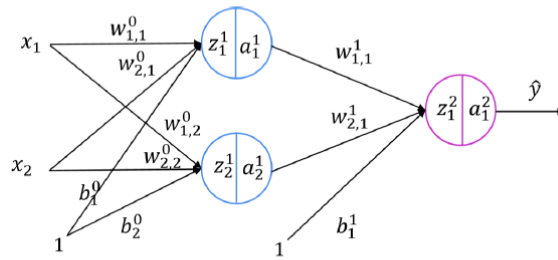
$$\frac{\partial E}{\partial a_1^2} \quad E = \frac{1}{2}(y - a_1^2)^2$$

$$\frac{\partial E}{\partial a_1^2} = \frac{1}{2} * 2 * (-1)(y - a_1^2) = (a_1^2 - y)$$

$$\frac{\partial a_1^2}{\partial z_1^2} \quad a_1^2 = \frac{1}{1 + e^{-z_1^2}} \quad \frac{\partial a_1^2}{\partial z_1^2} = \frac{-e^{-z_1^2}}{(1 + e^{-z_1^2})^2} = a_1^2(1 - a_1^2)$$

$$\frac{\partial z_1^2}{\partial w_{1,1}^1} \quad z_1^2 = w_{1,1}^1 a_1^1 + w_{2,1}^1 a_2^1 + b_1^1 \quad \frac{\partial z_1^2}{\partial w_{1,1}^1} = a_1^1$$

$$\frac{\partial E}{\partial w_{1,1}^1} = (a_1^2 - y) * a_1^2(1 - a_1^2) * a_1^1$$



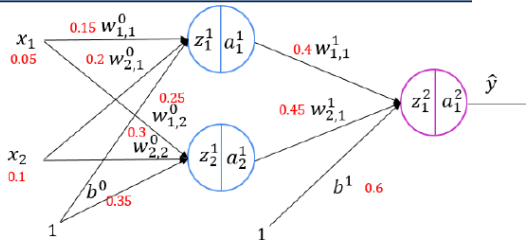
Backpropagation

$$\frac{\partial E}{\partial w_{1,1}^1} = \frac{\partial E}{\partial a_1^2} * \frac{\partial a_1^2}{\partial z_1^2} * \frac{\partial z_1^2}{\partial w_{1,1}^1}$$

$$\delta_{a_1^2} = \frac{\partial E}{\partial a_1^2} * \frac{\partial a_1^2}{\partial z_1^2}$$

$$\delta_{a_1^2} = (a_1^2 - y) * a_1^2(1 - a_1^2)$$

$$\frac{\partial E}{\partial w_{1,1}^1} = \delta_{a_1^2} * a_1^1$$



$$\delta_{a_1^2} = (0.7514 - 0.01) * 0.7514 * (1 - 0.7514) = 0.1385 \quad \frac{\partial E}{\partial w_{1,1}^1} = 0.1385 * 0.5933 = 0.0822$$

Cuánto un cambio en $w_{1,1}^1$ afecta a E

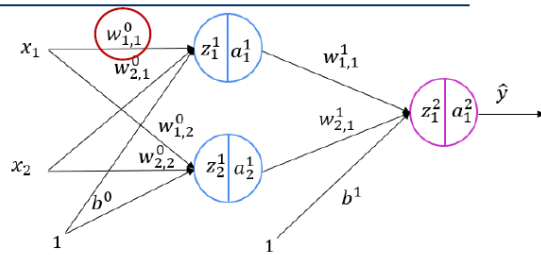
$$\text{supongamos } \eta = 0.5 \quad w_{1,1}^1(t+1) = w_{1,1}^1(t) - \eta \frac{\partial E}{\partial w_{1,1}^1} = 0.4 - 0.5 * 0.082 = 0.3590$$

Backpropagation

$$w_{1,1}^0(t+1) = w_{1,1}^0(t) - \eta \frac{\partial E}{\partial w_{1,1}^0}$$

$$\frac{\partial E}{\partial w_{1,1}^0}$$

$$E = \frac{1}{2} (y - \hat{y})^2 = \frac{1}{2} (y - a_1^2)^2$$



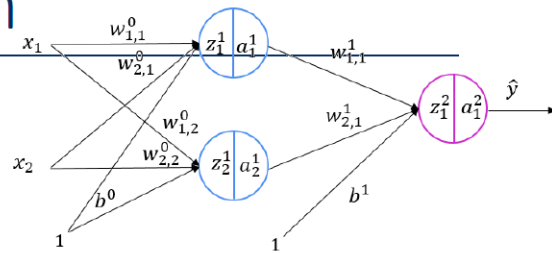
$$\frac{\partial E}{\partial w_{1,1}^0} \longrightarrow a_1^2 = \frac{1}{1 + e^{-z_1^2}} \longrightarrow z_1^2 = w_{1,1}^1 a_1^1 + w_{2,1}^1 a_2^1 + b^1 \longrightarrow a_1^1 = \frac{1}{1 + e^{-z_1^1}} \longrightarrow z_1^1 = w_{1,1}^0 x_1 + w_{2,1}^0 x_2 + b^0$$

$$\frac{\partial E}{\partial w_{1,1}^0} = \frac{\partial E}{\partial a_1^2} * \frac{\partial a_1^2}{\partial z_1^2} * \frac{\partial z_1^2}{\partial a_1^1} * \frac{\partial a_1^1}{\partial z_1^1} * \frac{\partial z_1^1}{\partial w_{1,1}^0}$$

Backpropagation

$$\frac{\partial E}{\partial w_{1,1}^0} = \frac{\partial E}{\partial a_1^2} * \frac{\partial a_1^2}{\partial z_1^2} * \frac{\partial z_1^2}{\partial a_1^1} * \frac{\partial a_1^1}{\partial z_1^1} * \frac{\partial z_1^1}{\partial w_{1,1}^0}$$

$\delta_{a_1^2}$



$$\frac{\partial z_1^2}{\partial a_1^1}$$

$$z_1^2 = w_{1,1}^1 a_1^1 + w_{2,1}^1 a_2^1 + b^1 \quad \frac{\partial z_1^2}{\partial a_1^1} = w_{1,1}^1$$

$$\frac{\partial a_1^1}{\partial z_1^1}$$

$$a_1^1 = \frac{1}{1 + e^{-z_1^1}} \quad \frac{\partial a_1^1}{\partial z_1^1} = a_1^1 * (1 - a_1^1)$$

$$\frac{\partial z_1^1}{\partial w_{1,1}^0}$$

$$z_1^1 = w_{1,1}^0 x_1 + w_{2,1}^0 x_2 + b^0 \quad \frac{\partial z_1^1}{\partial w_{1,1}^0} = x_1$$

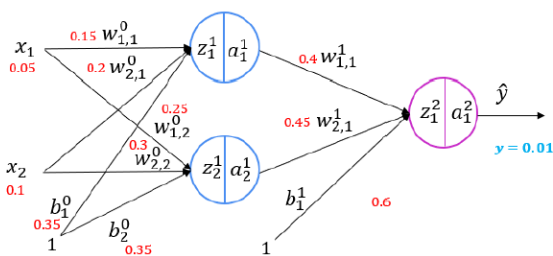
$$\frac{\partial E}{\partial w_{1,1}^0} = \delta_{a_1^2} * \frac{\partial z_1^2}{\partial a_1^1} * \frac{\partial a_1^1}{\partial z_1^1} * \frac{\partial z_1^1}{\partial w_{1,1}^0} = \delta_{a_1^2} * w_{1,1}^1 * a_1^1 * (1 - a_1^1) * x_1$$

Backpropagation

$$\frac{\partial E}{\partial w_{1,1}^0} = \delta_{a_1^2} * w_{1,1}^1 * a_1^1 * (1 - a_1^1) * x_1$$

$$\delta_{a_1^1} = \delta_{a_1^2} * w_{1,1}^1 * a_1^1 * (1 - a_1^1)$$

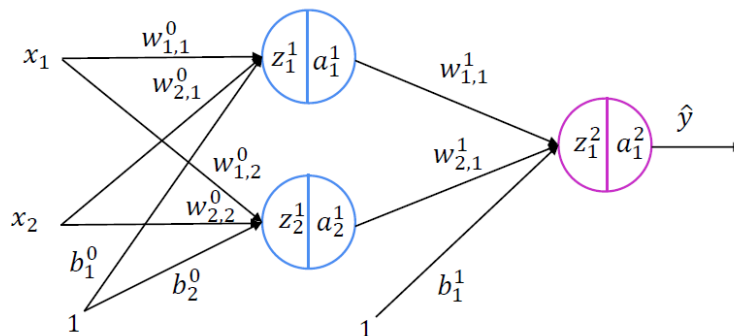
$$\delta_{a_1^1} = 0.1385 * 0.0965$$



$$\frac{\partial E}{\partial w_{1,1}^0} = \delta_{a_1^2} * \delta_{a_1^1} * x_1 = 0.1385 * 0.0965 * 0.05 = 6.6835e - 04$$

supongamos $\eta = 0.5$ $w_{1,1}^0(t+1) = w_{1,1}^0(t) - \eta \frac{\partial E}{\partial w_{1,1}^0} = 0.15 - 0.5 * 6.6835e - 04 = 0.1497$

Backpropagation



Matricialmente

Capas $l: 0 \dots L$ $A^0 = [x_1 \ x_2 \ 1]$ $Z^1 = [z_1^1 \ z_2^1]$ $A^1 = [a_1^1 \ a_2^1 \ 1]$

$$W^0 = \begin{bmatrix} w_{1,1}^0 & w_{1,2}^0 \\ w_{2,1}^0 & w_{2,2}^0 \\ b_1^0 & b_2^0 \end{bmatrix} \quad W^1 = \begin{bmatrix} w_{1,1}^1 \\ w_{2,1}^1 \\ b_1^1 \end{bmatrix}$$

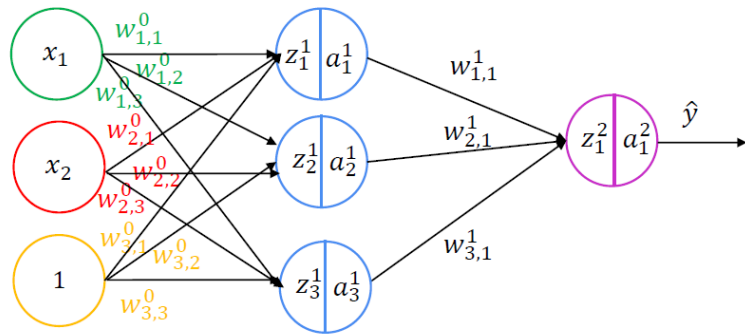
Forward propagation

Matricialmente

Capas $l: 0 \dots L$

$$W^0 = \begin{bmatrix} w_{1,1}^0 & w_{1,2}^0 & w_{1,3}^0 \\ w_{2,1}^0 & w_{2,2}^0 & w_{2,3}^0 \\ w_{3,1}^0 & w_{3,2}^0 & w_{3,3}^0 \end{bmatrix}$$

$$W^1 = \begin{bmatrix} w_{1,1}^1 \\ w_{2,1}^1 \\ w_{3,1}^1 \end{bmatrix}$$



Producto matricial

$$A^1 = \text{sigmoide}(A^0 * W^0)$$

$$A^2 = \text{sigmoide}(A^1 * W^1)$$

$$A^0 = [x_1 \quad x_2 \quad 1] \quad A^1 = [a_1^1 \quad a_2^1 \quad a_3^1]$$

$$A^2 = [a_1^2]$$

Backpropagation

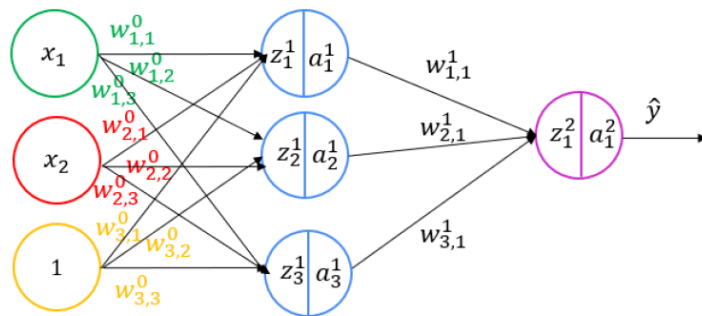
Capa 2

$$w_{1,1}^1(t+1) = w_{1,1}^1(t) - \eta \delta_{a_1^2} a_1^1$$

$$w_{2,1}^1(t+1) = w_{2,1}^1(t) - \eta \delta_{a_1^2} a_2^1$$

$$w_{3,1}^1(t+1) = w_{3,1}^1(t) - \eta \delta_{a_1^2} a_3^1$$

$$\delta_{a_1^2} = (a_1^2 - y) * a_1^2(1 - a_1^2)$$



Capa 1

$w_{1,1}^0(t+1) = w_{1,1}^0(t) - \eta \delta_{a_1^1} x_1$	$w_{1,2}^0(t+1) = w_{1,2}^0(t) - \eta \delta_{a_2^1} x_1$	$w_{1,3}^0(t+1) = w_{1,3}^0(t) - \eta \delta_{a_3^1} x_1$
$w_{2,1}^0(t+1) = w_{2,1}^0(t) - \eta \delta_{a_1^1} x_2$	$w_{2,2}^0(t+1) = w_{2,2}^0(t) - \eta \delta_{a_2^1} x_2$	$w_{2,3}^0(t+1) = w_{2,3}^0(t) - \eta \delta_{a_3^1} x_2$
$w_{3,1}^0(t+1) = w_{3,1}^0(t) - \eta \delta_{a_1^1}$	$w_{3,2}^0(t+1) = w_{3,2}^0(t) - \eta \delta_{a_2^1}$	$w_{3,3}^0(t+1) = w_{3,3}^0(t) - \eta \delta_{a_3^1}$
$\delta_{a_1^1} = \delta_{a_1^2} \cdot w_{1,1}^1 \cdot a_1^1 \cdot (1 - a_1^1)$	$\delta_{a_2^1} = \delta_{a_1^2} \cdot w_{2,1}^1 \cdot a_2^1 \cdot (1 - a_2^1)$	$\delta_{a_3^1} = \delta_{a_1^2} \cdot w_{3,1}^1 \cdot a_3^1 \cdot (1 - a_3^1)$

Backpropagation

Matricialmente
 Capas $l: 0 \dots L$

$$W^0 = \begin{bmatrix} w_{1,1}^0 & w_{1,2}^0 & w_{1,3}^0 \\ w_{2,1}^0 & w_{2,2}^0 & w_{2,3}^0 \\ w_{3,1}^0 & w_{3,2}^0 & w_{3,3}^0 \end{bmatrix} \quad W^1 = \begin{bmatrix} w_{1,1}^1 \\ w_{2,1}^1 \\ w_{3,1}^1 \end{bmatrix} \quad A^0 = [x_1 \quad x_2 \quad 1]$$

$A^1 = [a_1^1 \quad a_2^1 \quad a_3^1] \quad A^2 = [a_1^2]$

Capa 1

$$\delta_{a_1^1} = \delta_{a_1^2} \cdot w_{1,1}^1 \cdot a_1^1 \cdot (1 - a_1^1)$$

$$\delta_{a_2^1} = \delta_{a_1^2} \cdot w_{2,1}^1 \cdot a_2^1 \cdot (1 - a_2^1)$$

Capa 2

$$\delta_{a_1^2} = (a_1^2 - y) \cdot a_1^2 \cdot (1 - a_1^2)$$

$$\Delta^2 = [\delta_{a_1^2}] = (a_1^2 - y) \cdot (A^2)'$$

$$W^1(t+1) = W^1(t) - \eta (A^1)^T * \Delta^2$$

$$\delta_{a_3^1} = \delta_{a_1^2} \cdot w_{3,1}^1 \cdot a_3^1 \cdot (1 - a_3^1)$$

$$\Delta^1 = [\delta_{a_1^1} \quad \delta_{a_2^1} \quad \delta_{a_3^1}] =$$

$$= \Delta^2 * \begin{bmatrix} w_{1,1}^1 \\ w_{2,1}^1 \\ w_{3,1}^1 \end{bmatrix}^T \cdot [(a_1^1)' \quad (a_2^1)' \quad (a_3^1)']$$

$$\Delta^1 = \Delta^2 * (W^1)^T \quad \Delta^1 = \Delta^1 \cdot (A^1)'$$

$$W^0(t+1) = W^0(t) - \eta (A^0)^T * \Delta^1$$